

	Note !
	It is suggested that different coloured drums are used, which also helps identify the different purposes.

It is recommended to store larger quantities of debris outside of the factory in a clearly defined location away from other sources of ignition or materials which can burn.

Ensure the AlOx debris is covered to prevent direct contact with any rain

Use lids on all containers to reduce the source of oxygen required to support the fire (fire triangle) but ensure the lids are not sealed to prevent pressurising the container. Drill the side of the drum with a single 6mm hole.



Figure 5.2-18 Example of an external storage solution

	Note !
	The use of a metal container to store multiple containers encloses the risk and also prevents rain from entering the AlOx debris. The doors of the container should be welded open to allow good air flow through the container and also prevent the build-up of hydrogen gas.



Figure 5.2-19 Example of an external storage solution

	Warning !	
	Do not mix pure Aluminium metal and AlOx waste in the same containers	



Figure 5.2-20 Example of identified drums of different colours

Before any AlOx debris is stored in the 45 gallon steel drum, a 6mm hole is drilled in the side of it, to stop the potential of any pressure build up, when the lid is installed later.



Figure 5.2-21 Example of an operator drilling 6mm hole in the side of the container.



Figure 5.2-22 Maximum level of AlOx debris, 1/3 the total height of the drum

	Note !
	It is recommended that the storage drum is only filled to 1/3 full, and the level is below the hole drilled in the side of the drum



Figure 5.2-23 Example of a commercially available containment pallet

	Note !
	It is recommended that a steel containment pallet be used

Dedicated Vacuum Cleaner

It is recommended that an industrial rated vacuum cleaner be used to allow a high level of AlOx dust collection from the machine, when cleaning takes place. It should be rated for combustible dust.



Figure 5.2-24 Example of a suitable vacuum cleaner

If the AlOx dust is not collected, it can become airborne, every time the chamber is vented to atmosphere and contaminate other areas of the environment.

A suggestion of a suitable industrial vacuum supplier is <http://www.nilfiskcfm.com>.

	Warning !
	Empty to container after each cleaning cycle Do not allow for the collection level to be more than 1/3 by volume



Figure 5.2-25 Example of an appropriate identification label

	Warning !
	Ensure that the vacuum cleaner has a label attached that clearly identifies its purpose

Disposal of the AlOx Waste

	Warning !
	It is the user's responsibility to ensure the disposal company is fully aware of the risks associated with AlOx debris, as advised by Bobst Manchester Ltd.

	Note !
	The machine is interlocked not to allow the process to operate, if there is any issue with this gas dilution.

Spark Proof Tooling

To reduce the possibility of a fire occurring when cleaning, it is recommended that Copper/beryllium tools are used.

A source of this tooling is:-

<http://www.ngkmetals.com/index.cfm/m/55/fuseaction/store6Catalog.oneCategory/Scrapers>

	Warning !
	DO NOT USE ANY TOOLS THAT COULD CAUSE A SPARK

Personnel Protection Equipment (PPE)

It is advised that operators use personal protection equipment in accordance with the standards specified in Appendix E of this manual and follow the guidance provided by suppliers of any cleaning substances. The following table details the parts of the body at risk during cleaning and recommended clothing to address these risks.

Notes!	
	Eyes - Safety goggles (not glasses) (ANSI Z87.1-2003-6.2.1) capable of protecting the user from dust & projectiles should be worn by all personnel on the machine when cleaning is taking place.
	Hands – Gloves capable of protecting the user from abrasion, temperature extremes, cuts and punctures, impact, chemicals.
	Lungs – Respirators (EN 149:2001) capable of protecting the user from breathing in contaminants in the air are required to be worn when cleaning the machine.
	Body – Flame-retardant overalls or boiler suits (ISO 14116:2008) capable of protecting the user from heat and potential fire.
	Feet - Safety boots or shoes with protective toecaps and penetration-resistant, mid-sole capable of protecting the user from sharp or heavy objects.

Bobst Manchester Ltd. recommends that PPE clothing be provided to the following standard, ISO 14116:2008 for use by operational staff when operating this equipment, when the chamber is open, and aluminium oxide is present

ISO 14116:2008 specifies the performance requirements for the limited flame spread properties of materials, material assemblies and protective clothing in order to reduce the possibility of the clothing burning and thereby itself constituting a hazard.

Additional requirements for clothing are also specified in appendix E of this document.



Figure 5.2-26 Recommended PPE equipment

5.3 Hardware

To allow for this process to be incorporated onto the machine, extra hardware has been incorporated within it. This equipment is described briefly in this section of the manual.

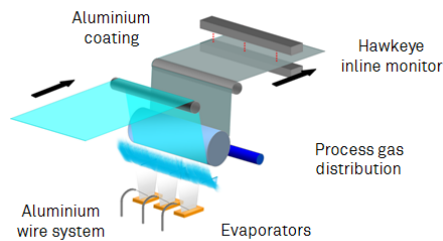


Figure 5.3-1 Process variables

Hawkeye Inline Monitor System

As part of this system, the Hawkeye inline monitor system is supplied to allow measurement of very thick layer of the aluminium coating (>2.2 OD).

This operation, calibration, etc. of device is covered by other sections of this manual.



Figure 5.3-2 Hawkeye reporting System display screen



Figure 5.3-3 Example of the Hawkeye system with the receiver probes removed

Aluminium Wire Feed System

The same aluminium wire feed system is used to provide the aluminium for the process, as if standard aluminium metallizing process. [See chapter Section 3.0 Aluminium Evaporation Process on page 77 for details.](#)

Gas Delivery System

The process gas delivery system consist of the following equipment:-

- Central process gas distribution/connection panel
- External (outside the vacuum chamber) process gas distribution lines

- Mass flow controllers (MFCs), including isolation solenoids
- Internal (Inside the vacuum chamber) process gas distribution lines.

This equipment is shown on the gas schematic O2 drawing for each contract

	Caution !
	The process gas supply to the distribution panel should be regulated to feed a pressure of 4-5 bar (G). It is recommended that a pressure relief valve is installed in the supply system to allow relief at 7 Bar.

Central Process Gas Distribution/connection Panel

This is the panel that is used to monitor and distribute the gas supply from the end users connection.

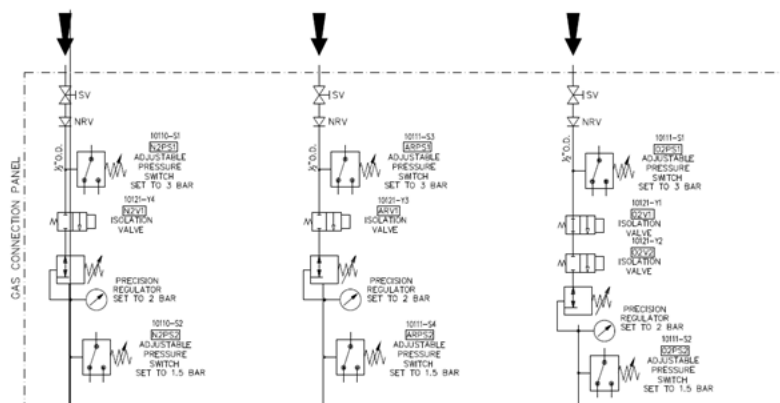


Figure 5.3-4 Gas panel components and their default settings

	DANGER !
	DO NOT MODIFY THIS SYSTEM OR THE SETTINGS OF THE INTERLOCKING EQUIPMENT

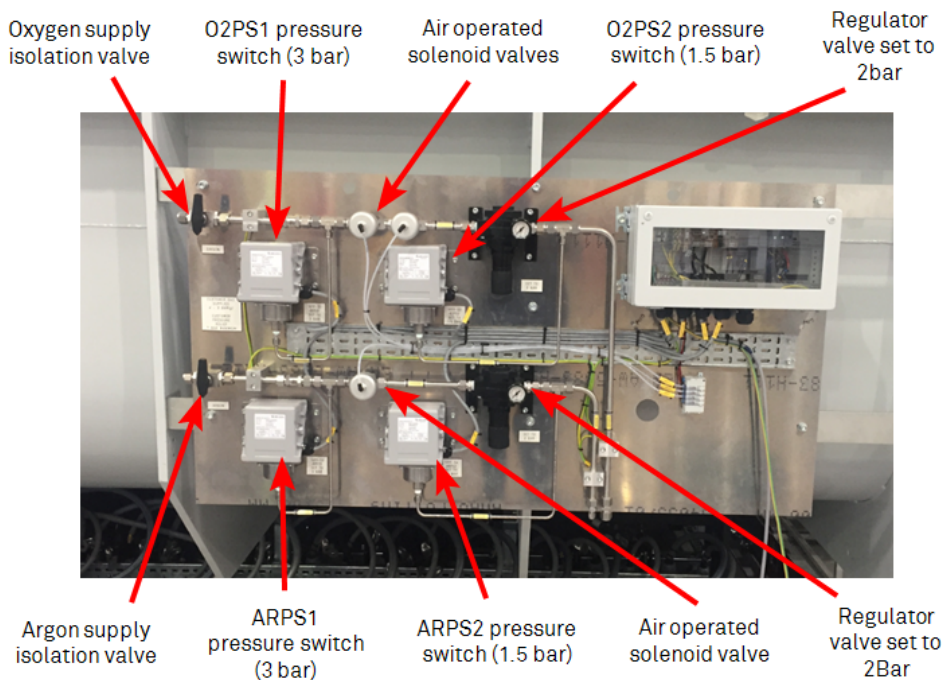


Figure 5.3-5 Gas distribution panel

External Process Gas Distribution Lines

Clearly identified stainless steel rigid and flexible pipework is used for the gas distribution system.

	DANGER ! DO NOT MODIFY THIS GAS DISTRIBUTION PIPEWORK SYSTEM OR SUBSTITUTE NON EQUIVALENT PARTS
	Warning ! DO NOT ATTEMPT TO DISMATE THIS DISTRIBUTION PIPEWORK WITHOUT ENSURING THAT THE PROCESS GAS HAS BEEN REMOVED.
	Note ! It is recommended that this pipework be checked on a regular timescale for damage and leakage.

Mass Flow Controllers (MFCs)/Isolation Solenoids

These individual controllers are used to regulate the amount of process gas being delivered into the evaporation area of the machine, for the Alubond™ process to take place.

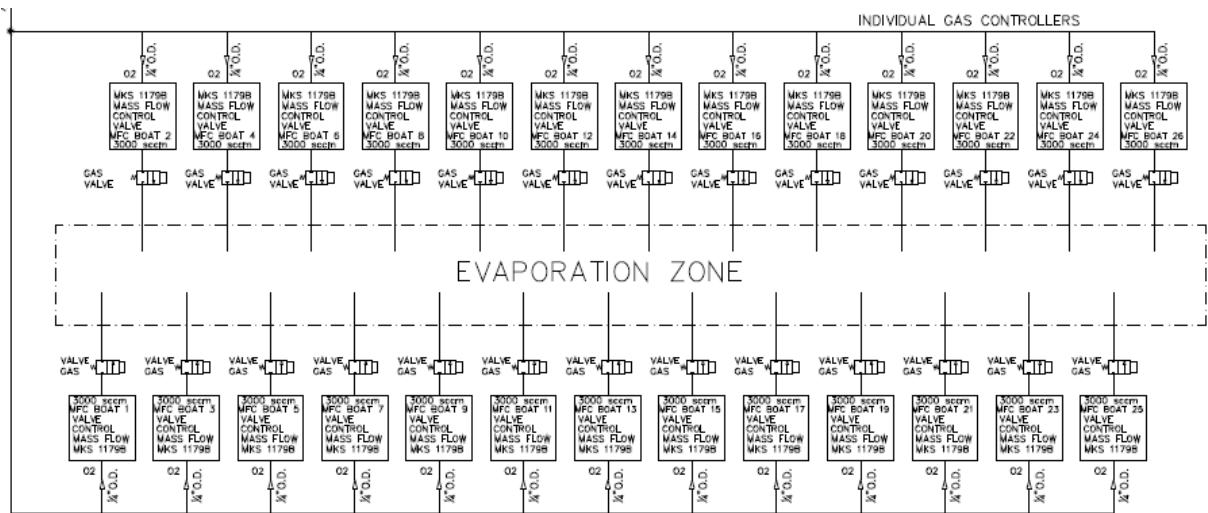


Figure 5.3-6 MFC components

The MFC controllers are located at a high level on the winding cart. These are controlled via the profibus communications network, as indicated by the “purple” cables.



Figure 5.3-7 MFC controllers

Internal Process Gas Distribution Lines.

Flexible lines are used within the chamber to deliver the process gas to silicone cups that match up with the coating shield, for the final routing of the gas into the evaporation area.



Figure 5.3-8 Silicone cups

	DANGER !
	DO NOT USE FACTORY COMPRESSED AIR SYSTEMS TO CLEAN ANY OF THESE PARTS. ONLY USE DRY NITROGEN

When the coating shield is moved into position for normal operation, the silicone cups align with flat surfaces on the shield to allow the distribution of the gas without leakage.

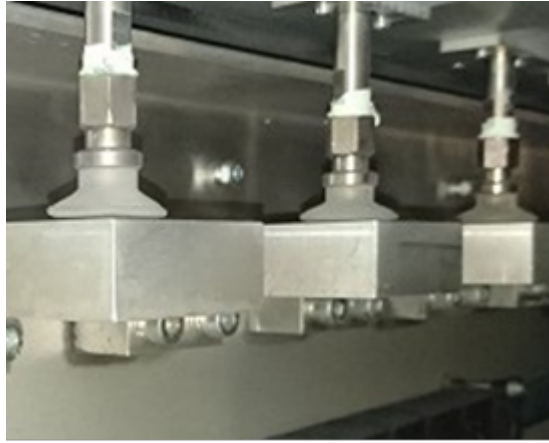


Figure 5.3-9 Cup to shield interface

Holes pass through the coating shield to mallow distribution of the reactant gas into the aluminium vapour cloud.



Figure 5.3-10 Gas delivery points on the coating shield

Over a period of time, possible blockage of these openings may occur. The adaptor and gas delivery line can be removed to allow cleaning to take place.

The distribution pipe can be removed, by removing the two cap-heads shown



Figure 5.3-11 Example of an operator removing a gas feed block.



Figure 5.3-12 Example of an operator removing gas distribution pipework.

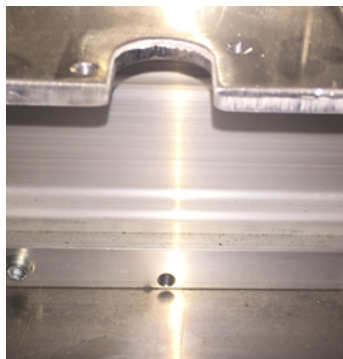


Figure 5.3-13 Hole in the shield that can be cleaned



Figure 5.3-14 Gas feed distribution pipework

Shield Temperature monitoring

There are a number of thermocouples attached to the shielding located between the evaporation area and the chamber wall. The purpose is to monitor the temperature and to warn the user, if there is a rise in temperature, as the machine is vented. This increase could be caused by a reaction between the non-reacted Alox coating and the air entering the chamber.

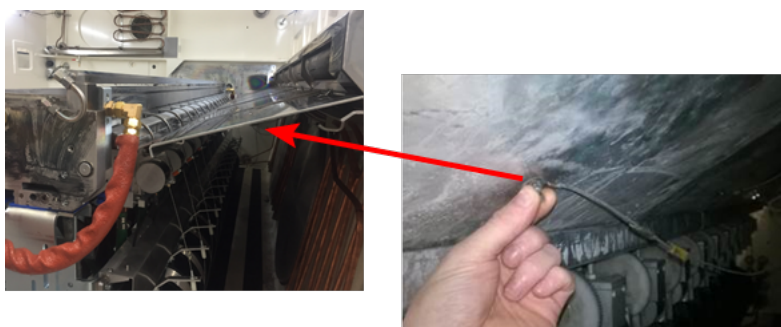


Figure 5.3-15 Location of the over temperature thermocouples

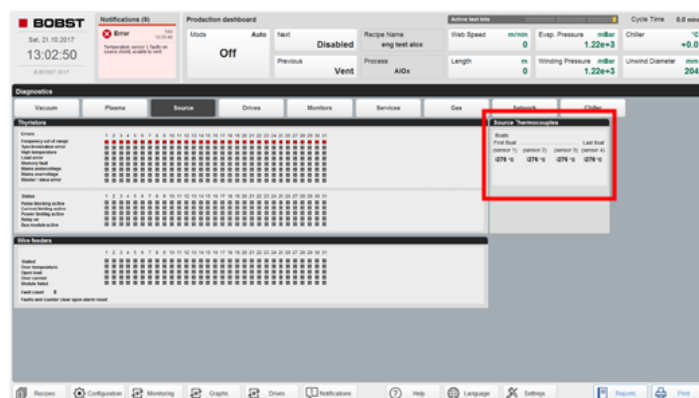


Figure 5.3-16 DIAGNOSTICS-SOURCE HMI Screen showing temperature feedback.

	DANGER ! DO NOT OPERATE THE MACHINE FOR THIS TYPE OF PROCESS WITHOUT THIS EQUIPMENT IN PLACE
---	--

	Note ! If the temperature reaches 250C on any of the thermocouples, the machine will stop the venting sequence.
---	---

Modified Chamber Venting System Components

An extra venting valve is installed onto the chamber and its purpose is to allow a more controlled slower vent of air into the chamber to take place.

It has been named VV10C on the Vacuum HMI, and has a smaller internal cross section, which slows down the air that can pass through it.

The installation and control of it will extend the venting sequence.

Cobra 630 Pump Admittance System

Extra components are added to the pump's air admittance system to ensure that air is flowing into the "dilution" port on the side of the pump body.



Figure 5.3-17 Standard pump (left) and modified pump (right)

Cobra 630 pump Exhaust System

As part of the installation of the exhaust ducting, flow monitoring sensors will inform the machine, if the exhaust system is operating correctly.

Baratron Gauge – Evaporation Zone

To allow for accurate pressure measurement in the evaporation zone, when the process gas is supplied, a Baratron gauge is installed to provide accurate feedback.

The pressure measurement is shown on the VACUUM HMI, as "BAR" in the evaporation area of the chamber.

Evaporation Source Height

To allow for full reaction in the vapour cloud the evaporation source is lowered and an extra cooled section is added to the top of it.

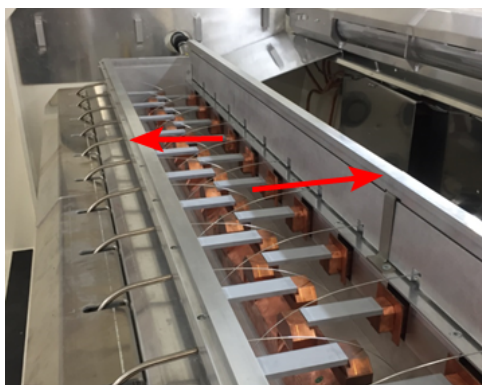


Figure 5.3-18 Example of the evaporation source with the extra cooled components



Figure 5.3-19 Example of the evaporation source without the extra cooled components

5.4 Machine Set Up

Based upon practical experience, the following areas of the machine, will need to be checked/setup prior to running this process.

These checks/actions should be extra to the normal machine set up actions.

Gas Bottles/supply

Check that gas process delivery system is available and that the isolation valves are open

Coating shield outlets

	Note ! To allow gas flow to take place, remove the blanking screws before attempting to run this process.
---	---

Remove the blanking plugs that are located in the gas delivery ports.



Figure 5.4-1 Brass screws located in the coating shield

Visually check that all the openings on the coating shield are clear of any aluminium build up. Use a small brass brush to clean the openings, if necessary.



DANGER !

Do not use any lubricants in this area, as potentially, they could react with the process gas and cause an explosion.



Figure 5.4-2 Example of a suitable brass brush for cleaning the threads.

Check the area between the flexible cup and the mating surfaces for obstructions/debris, and clean them.

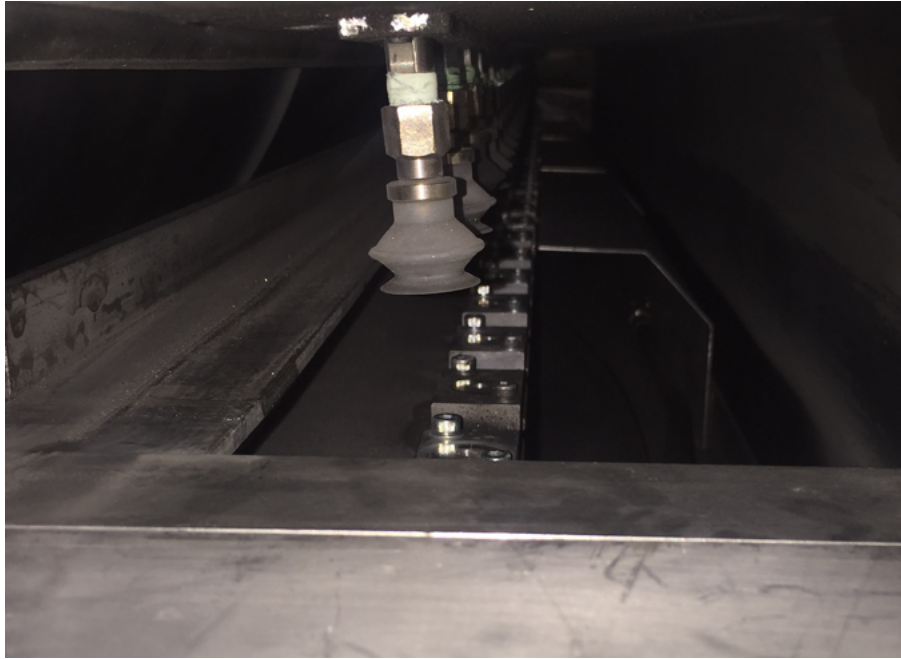


Figure 5.4-3 Area between the cup and the mating surface.

Coating Shield – Sine Waves

Ensure that no sine waves are used and blanking plates are in location. [See chapter 9.10 Coating Shield System on page 441 for details.](#)

Exhaust and Circulation Systems

Ensure that the systems are operational.

Aluminium Wire Placement on Evaporators

Ensure that the contact point on every evaporator is middle of the evaporator

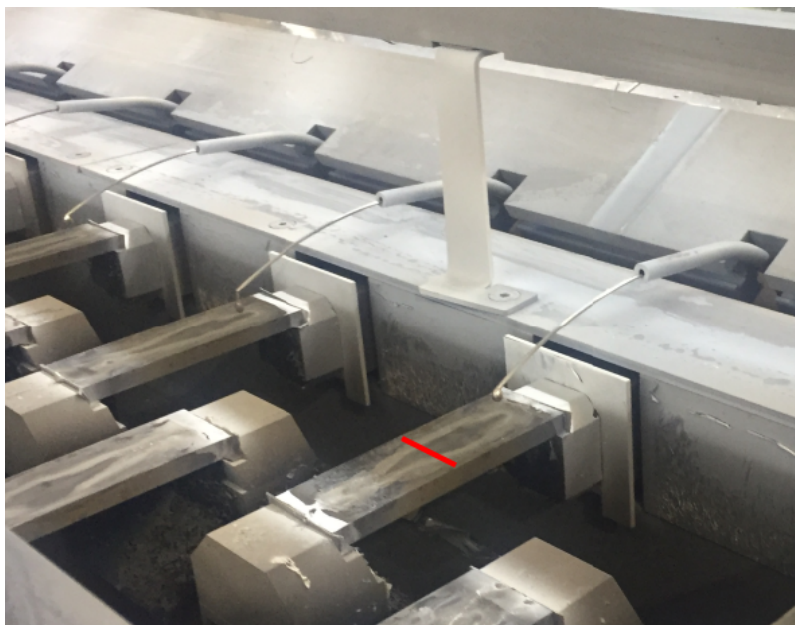


Figure 5.4-4 Aluminium contact point at 50% of the evaporator length

Recipe Page Setup

Select the RECIPES HMI screen

1. Set the chiller set-point to -5C
2. Set the “Pre-gas target” to 0.3 OD higher than the product target.
3. Select the process type to Alubond, select the control monitor as shown below

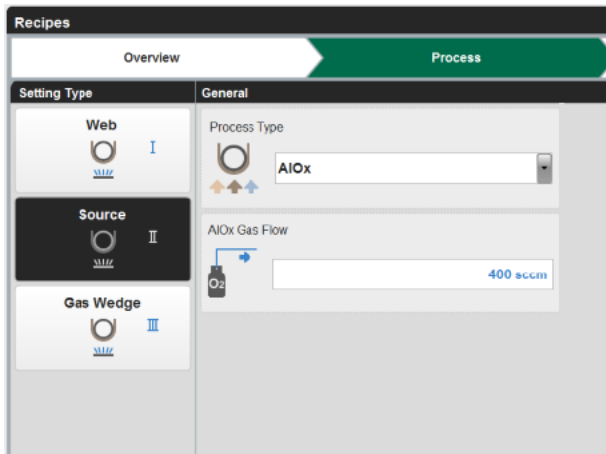


Figure 5.4-5 HMI RECIPES- PROCESS screen

4. Set the targets as shown below for 600m/min operation

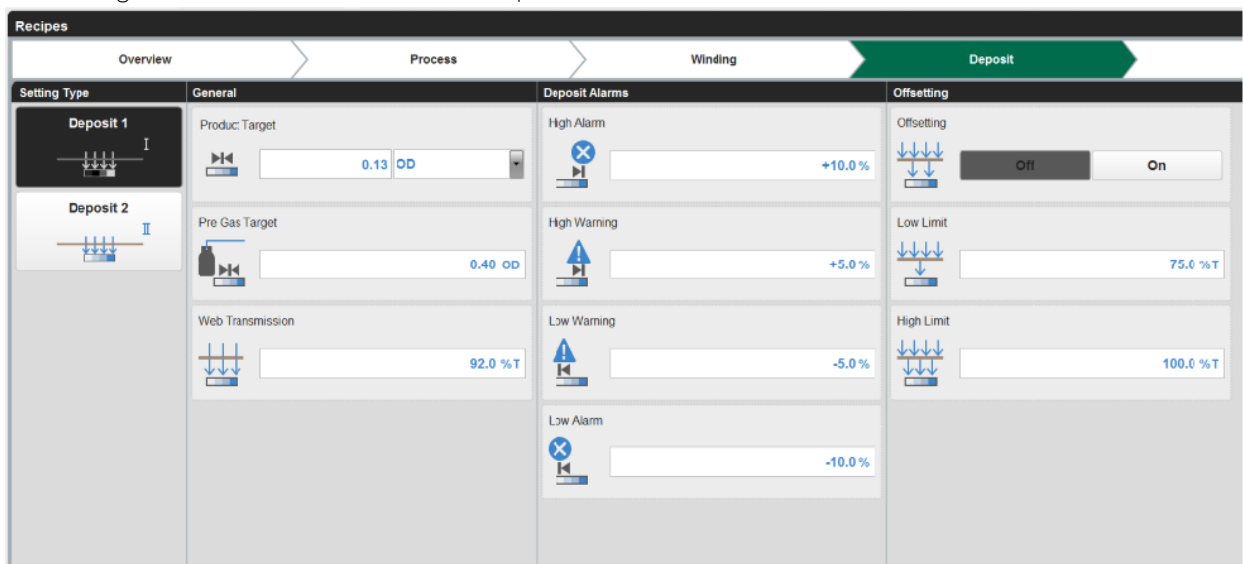


Figure 5.4-6 HMI RECIPES- DEPOSIT screen

5.5 Process Guidance - Conditions

This section of the manual is written to explain the interfacing HMI screen for this process.

When the Alubond™ process is selected from the setup screen, the SOURCE PROCESS screen will change from the standard aluminium metallizing screen.

It can be seen that initially, the “Pre-gas” OD target (2.70) will be set, as shown below.



Figure 5.5-1 PROCESS HMI interface screen

The machine should be prepared and the steps described in section 9.6 of this manual be followed, until the evaporator power (BOATS ON) is applied and the auto heat up cycle has completed.

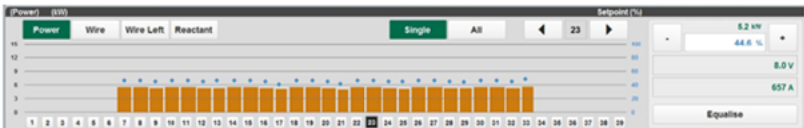


Figure 5.5-2 PROCESS HMI interface screen – evaporators at temperature

1. The power set points for the heaters should be initially increased by 10% to allow for full wetting out of the evaporators, using the ALL Channel button and the increase arrow. [See chapter 3.5 Evaporator Normal Running Mode on page 90 for details.](#)



Figure 5.5-3 Example of a heated evaporation source

- 2. Select the wire feed rate for wetting out. It is recommended that this rate is set at 50% of the process wire speed.
- 3. Start the aluminium wire feeders by pressing the WIRE on button.

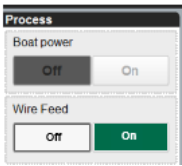


Figure 5.5-4 MONITORING HMI interface screen- BOAT POWER & WIRE FEED control.

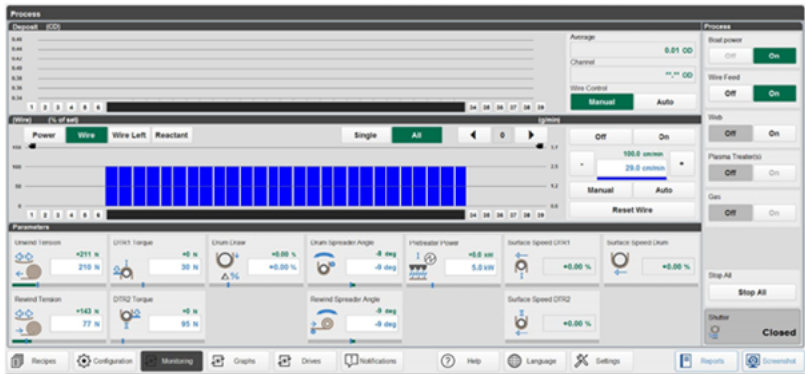


Figure 5.5-5

Figure 5.5-6 MONITORING HMI interface screen showing aluminium wire being fed

- 4. Check that the wire is feeding correctly and full cavity use is taking place. If the evaporator coverage is not full, reduce the power to the individual evaporators to correct this, until the darkened area on the evaporator surface is wide.

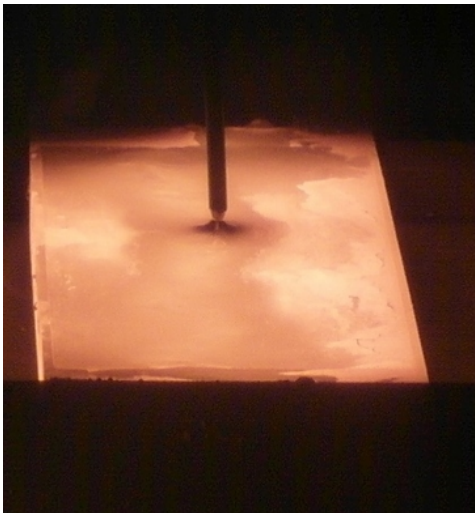


Figure 5.5-7 Example of an evaporator with full coverage

- 5. Enable the plasma gas and power, if this has not already been carried out
- 6. Start the web running and allow the shutter to open and stable conditions to be established.
- 7. Modify the evaporator power set-points to establish the uniformity of the “pre-gas” target of 0.4 OD.

	Note !
	Check the average OD value, and bring inline the individual channels that may be out of alignment

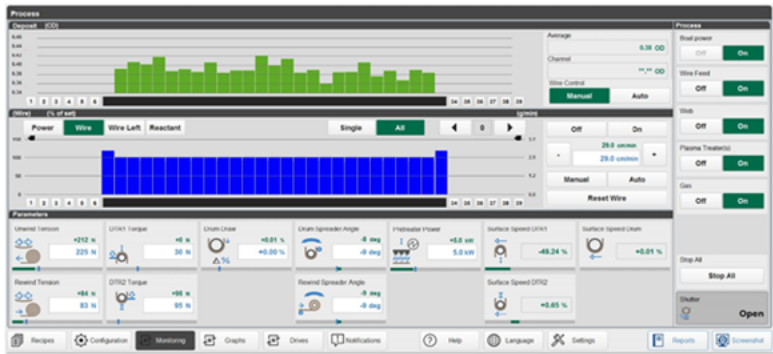


Figure 5.5-8 PROCESS HMI interface screen showing deposition feedback of the “pre-gas” deposition
At this time, the coating on the substrate, viewed in the machine will be grey.

- 8. Check that all readings are within +/-10% of the required deposition level.
- 9. Press the GAS ON button to start the process gas flow at the reactant gas at the ALUBOND GAS FLOW set point and the interface will change, to display the controlled gas output levels.



Figure 5.5-9 PROCESS HMI interface screen showing deposition feedback without auto control

- 10. Wait for 5-10 seconds
- 11. Select the REACTANT AUTO button.

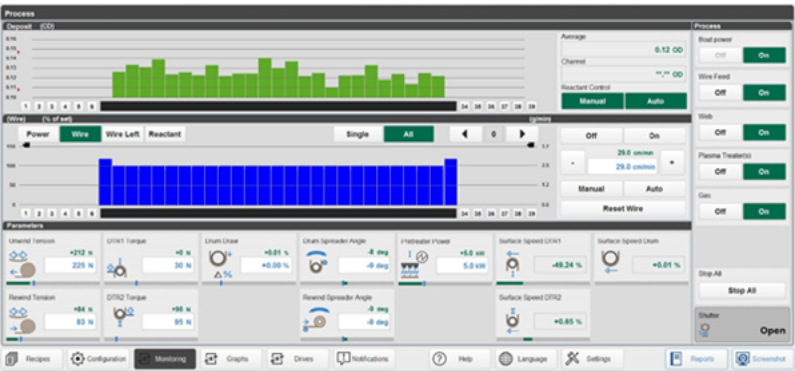


Figure 5.5-10 PROCESS HMI interface screen showing deposition and wire feeders
The target OD will change to the “Product target” on the deposition area of the process screen.
The control of the gas will be automatic, and this will be shown by the individual output REACTANT their outputs.
The REACTANT Control button, will flash until in full control and then will be shown solid green.



Figure 5.5-11 PROCESS HMI interface screen showing deposition and reactant gas flow.

12. Check the evaporators still have full cavity and make adjustments to the evaporator power levels as required.
As the unwind roll approaches the core, the machine will automatically start to auto slowdown sequence.

	Note!
	In comparison with the standard aluminium process, there are some extra steps to ensure that all process gas is isolated and that the supply lines are emptied of any gas, back to the gas distribution panel.

Chamber Venting

The vent sequence for this process, is controlled to be at a lower rate of flow into the chamber.

This is required to reduce the possibility of a reaction between the AlOx debris on the coating shields, etc. when the air is vented into the machine.

While the possibility of this occurring is low, if cleaning/operational procedures are followed, there is still a potential hazard.

From experience the area which collects the most AlOx coating is the shields between the evaporation source and the chamber wall.

These have temperature sensors attached to them to monitor the temperature.

If there is a reaction, when venting the chamber, and the monitored shield temperature rises above 250C, the venting sequence will stop.

	DANGER !
	DO NOT OPERATE THE MACHINE FOR THIS TYPE OF PROCESS WITHOUT THIS TEMPERATURE MONITORING EQUIPMENT IN PLACE

5.6 Cleaning Methods

This section of the manual replaces part of the standard methods used when cleaning standard aluminium debris, as the AlOx debris is potentially reactive, where standard aluminium coatings are not.

The following recommendations are made based upon the experience of our staff and customers, who kindly allowed photographs of their cleaning methods to be included in this document.

	Note !
	It is advised that all users are made aware of the hazards of AlOx and have read and understood all the relevant sections of this document

Personal Protective Equipment (PPE)

It is advised that operators use personal protection equipment in accordance with the standards specified in Appendix E of this manual and follow the guidance provided by suppliers of any cleaning substances. The following table details the parts of the body at risk during cleaning and recommended clothing to address these risks.

Note!	
	Eyes - Safety goggles (not glasses) (ANSI Z810.1-2003-6.2.1) capable of protecting the user from dust & projectiles should be worn by all personnel on the machine when cleaning is taking place.
	Hands – Gloves capable of protecting the user from abrasion, temperature extremes, cuts and punctures, impact, chemicals.
	Lungs – Respirators (EN 149:2001) capable of protecting the user from breathing in contaminants in the air are required to be worn when cleaning the machine.
	Body – Flame-retardant overalls or boiler suits (ISO 14116:2008) capable of protecting the user from heat and potential fire.
	Feet - Safety boots or shoes with protective toecaps and penetration-resistant, mid-sole capable of protecting the user from sharp or heavy objects.

Operator Hand Cleaning Tools

It is recommended that good quality-cleaning tools be provided and used in the cleaning procedures described in this section of the manual to provide the most efficient use of the equipment and minimum damage.

Stiff beryllium copper blade scrapers for cleaning surfaces contaminated by aluminium oxide deposits

These tools and explosions in environments where flammable residues are present. While many nonferrous alloys meet non-sparking standards, only beryllium copper has the superior strength and hardness, and should be used



Figure 5.6-1 Recommended cleaning tools for cleaning machines contaminated with AlOx debris

	Warning !
	Do not use steel or other tools that do not meet the specifications of the tools shown above. Ensure that a suitable fire extinguisher is available at the location of cleaning Follow the guidelines given in section 5.2 in regards to AlOx storage
	AlOx Debris Cleaning Guidance Notes !
	• After withdrawing the winding cart, leave the AlOx debris to finish reacting prior to cleaning the machine (Complete roll change over first) • Regular cleaning after each process cycle of the AlOx debris in the machine will minimize the risk of a metal fire on the machine • Thin deposited coatings have less potential for a reaction. • Be more vigilant when cleaning less reacted films (Black coatings) • Use the collection and storage equipment that is clearly identified for this use

Figure 5.6-2



Figure 5.6-3 Examples of an operator cleaning Alox deposits from the coating shield and source area on a vacuum metallizer



Note !

The following is a duplication of some of sections 6.2 and 10.1 of this manual

Vacuum Cleaner

It is recommended that an industrial rated vacuum cleaner be used to allow a high level of AlOx debris/dust collection from the machine, when cleaning takes place. It should be rated for combustible dust.

If the dust is not collected, it can become airborne, every time the chamber is vented to atmosphere and contaminate other areas of the environment.



Figure 5.6-4 Example of a suitable industrial type vacuum cleaner



Warning !

Indicates loss of life, severe personal injury, or substantial machinery damage will result if proper precautions are not taken.

AlOx Debris Waste Storage Container

The waste from the process, is required to be removed from the machine and removed from site through a recycle company, as it is still of a high grade.

This includes multi-layer pieces and dust



Figure 5.6-5 Example of a suitable storage drum for the AlOx debris

	Warning ! AlOx dust/debris should be stored in sealable steel containers to stop the contents becoming damp prior to disposal offsite. These should have a vent hole as described in section 6.2 of this manual
---	--

The steel drums should have clear identification of the contents that are stored within it and the lid should be applied at all times, when access is not required.



Figure 5.6-6 Example of appropriate labelling on the steel container

**Note !**

The use of wire brushes and pneumatic tools is discouraged, as they create airborne dust particles.

A cart is provided to be placed under the coating shield area to minimise the contaminated area when cleaning the debris from the shield



Figure 5.6-7 Example of a cleaning cart

**Warning !**

Remove all AlOx debris at the end of each cleaning cycle from the cleaning cart

A smaller container is used at the machine to place the AlOx debris in.



Figure 5.6-8 Example of a small AlOx container

**Warning !**

Empty the container at the end of each cleaning cycle



Note !

It is suggested that different coloured drum are used, which also helps identify the different purposes.

It is recommended to store larger quantities of AlOx debris outside of the factory in a clearly defined location away from other sources of ignition or materials which can burn.

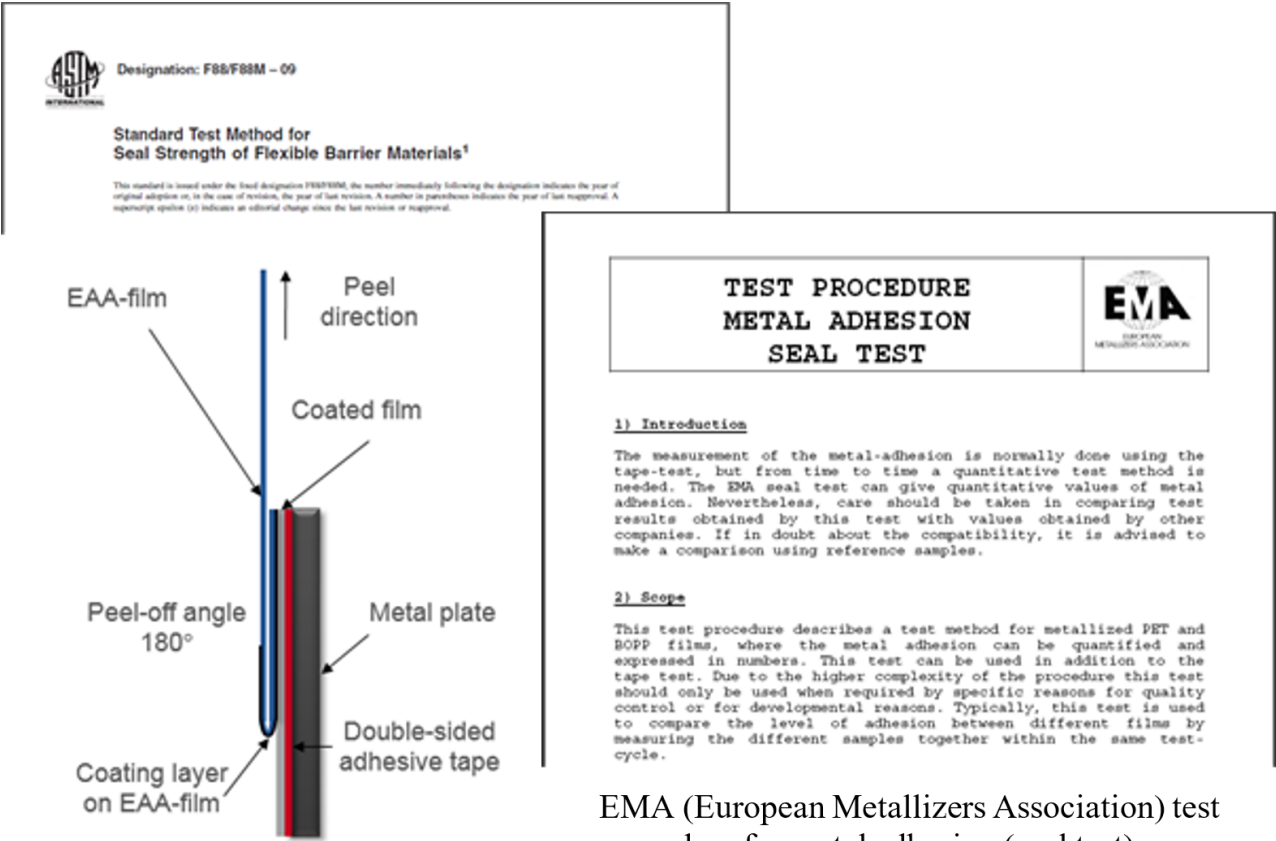


Warning !

Do not mix Aluminium metal and Alox waste in the same waste storage containers

5.7 Testing of Processed Material - Coating Adhesion

Adhesion/bond strength (general) -ASTM F88



Metal adhesion - EMA test method

Figure 5.7-1 Diagram showing test method

Laminate/adhesive Bond Strength – Standards, Testing Methods and Equipment

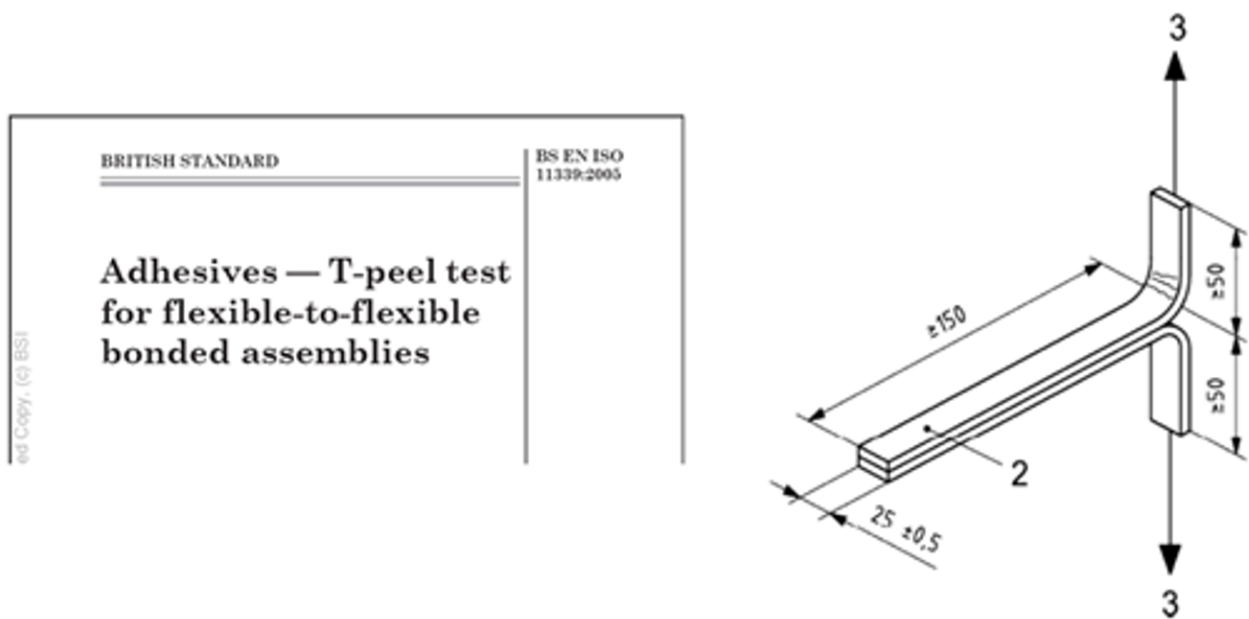


Figure 5.7-2 Diagram showing test method



Figure 5.7-3 Integral E100 EAA film details label

5.8 Process Start-Up/Operation/Shutdown Check Sheet

Bobst Manchester are providing the sheet on the page overleaf to be used as a template for the customer use.

A soft copy is provided to allow the customer to create further copies.

DAILY & WEEKLY CHECKS WHEN RUNNING THE ALUBOND PROCESS



A designated person must tick off checks twice daily as they are completed and initial the column. Checks performed weekly are greyed out for all apart from Monday. These checks must be performed at least once a week. These sheets are to be kept in a ring binder near the machine.

Week commencing: _____

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Physical/Visual checks to be carried out prior to running AIOx process							
1 Check process gas cylinders pressures/contents are sufficient for the process cycles intended for that day.							
2 Check main process gas cylinder isolation valves are open and supply is available to the gas distribution panel.							
3 Check main process gas isolation valves on the top of the distribution panel are open.							
4 Check that the vacuum pump exhaust fan extraction system is powered up and healthy indication on the monitoring devices is good							
5 Check that pit ventilation forced circulation fan is powered up.							
6 Inspect the oxygen analyser for to the pit area under the machine, and ensure that there is no alarm condition, and that the oxygen concentration is within the required limits.							
7 Check the quantity of the AIOx debris in the collection containers and empty them (more than 25% full).							
8 Ensure correct PPE is available for handling/cleaning AIOx is available							
9 Check all relevant fire fighting equipment is available and does not indicate that it has been used.							
10 Ensure that a clear path is available for removal of AIOx debris is available, and there are no obstructions.							
11 Check that the temperature monitoring thermocouples are in location.							
12 Check that the correct type cleaning tools are available							
Operator Initials:							
Supervisor Initials:							
Visual/physical checks to be carried out at the end of the AIOx process running period.							
1 Check main process gas cylinder isolation valves are closed.							
2 Check main process gas cylinder isolation valves are closed and supply is isolated from the gas distribution panel.							
3 Ensure that the all machine areas have been cleaned of any AIOx deposition.							
4 Empty the contents of the vacuum cleaner of any AIOx debris.							
5 Ensure that there is no AIOx debris is left anywhere but in the dedicated collection containers.							
6 Remove all containers with AIOx debris in them to external storage area.							
Operator Initials:							
Supervisor Initials:							

Note: This is an example of the typical checks which can be carried out to ensure a safe operating environment.

SECTION 6.0 PLASMA TREATER SYSTEMS

The Plasma Treater process option, including hardware is supplied with this machine. This section of the manual describes the Plasma Treater System in the following chapters:

- [6.1 Plasma Treater Process on page 286](#)
- [6.2 Safety Considerations on page 288](#)
- [6.3 Producing the process environment to allow the process to take place on page 289](#)
- [6.4 Plasma System Power Supply on page 298](#)
- [6.5 Plasma Process Setup - HMI on page 304](#)
- [6.6 Water Cooling and HMI Interfacing on page 307](#)
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6.1 Plasma Treater Process

Web based substrates may require surface preparation or chemical modification to enhance the adhesion of the deposition layer.

Plasma can be used to prepare surface of a substrate before the coating is applied. This is called plasma pre-treatment and can provide the following benefits:

- Improved adhesion of the coating
- Smooth coating surface of the substrate
- Enhance moisture and gas barrier

The substrate material will influence what surface treatments are necessary to prepare them for coating.

What is Plasma?

A plasma is a state of matter that contains enough electrically charged particles to affect its electrical properties and behaviour.

In an ordinary gas, each atom is electrically neutral. A gas becomes a plasma when the addition of energy causes a significant number of atoms to release some or all of their electrons. The remaining parts of those atoms are left with a positive charge and the detached negative electrons are free to move about. Those atoms and the resulting electrically charged gas are said to be ionized.

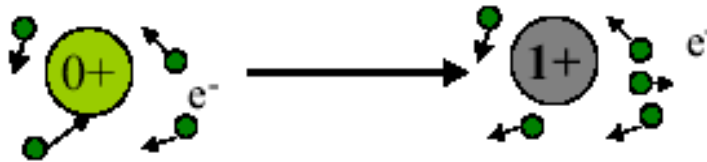


Figure 6.1-1 The effect of energy on an atom

When enough atoms are ionized to significantly affect the electrical characteristics of the gas, it is a plasma.

In plasma coating applications positive ions are generated by collisions between neutral particles and energetic electrons. The resulting plasma can be manipulated using a magnetic field.

In-Chamber Plasma Treatment

This machine uses a plasma system with an AC magnetron source that can be used with reactive gas blends.

This supplies high-energy electrons in sufficient quantities to generate heavy ions and additional electrons from the gas blend to bombard the surface of a substrate.

The ions and electrons are constrained in a ring shaped magnetic field above the generator and the substrate is passed through the plasma.

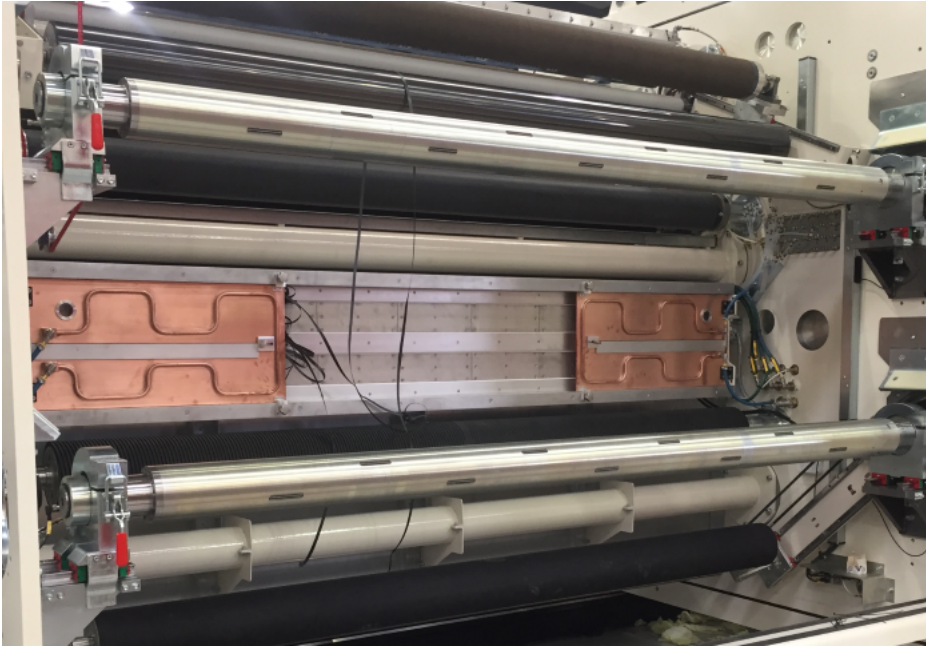


Figure 6.1-2 Example of a pre-treater plasma unit in a vacuum chamber

6.2 Safety Considerations

	Warning ! Further precautions for machines fitted with a plasma treater.
 	Any persons who have active or passive health aids upon their person must stay away from the equipment, when it is in operation. This is because the plasma power supply can create electromagnetic fields when operated. The risk is minimised because the electrodes are located inside the vacuum chamber and is only energised when it is closed and under a vacuum condition.